Day 7: ARP, MAC Vendor Codes and TTL

Level 1, Month 1, Networking foundations

What I learned

- ARP lets a device find the MAC address that belongs to an IP address on the same network.
- An ARP request is sent to the broadcast MAC `ff:ff:ff:ff:ff:ff` (48 bits, all 1), because the sender does not know the owner yet. The reply goes back to the sender only.
- The first 3 pairs of a MAC address identify the vendor. In my lab, `00:0c:29` means VMware, and Wireshark shows it as `VMware_...`.
- `arp -a` shows the ARP table of my computer. `dynamic` means learned by asking, `static` means built in (such as the broadcast).
- TTL lives in the IP header, not in the MAC address. It goes down by 1 at each router, and the packet is dropped at 0.

TTL initial values (a clue, not a proof)

System	Initial TTL
Windows	128
Linux, Android, iPhone, Mac	64
Cisco routers	255

- `TTL=64` means "not Windows". It can be Linux, Android, iPhone or Mac.
- Routers on the way = initial TTL - TTL received. With `TTL=117`: the system is probably Windows, and 128 - 117 = 11 routers.

Lab: ARP on my own virtual network

1. Started the machines in VMware (network type Host-only, `192.168.1.0/24`).
2. `ping 192.168.1.10`, then `arp -a`: the MAC next to `192.168.1.10` matched the one in my DNS project report.
3. Captured on `VMware Network Adapter VMnet1` with the display filter `arp`, then ran `ping 192.168.1.30`.

What I saw in the capture:

No.	Info	Meaning
1	Who has 192.168.1.30? Tell 192.168.1.1	Request, sent to broadcast
2	192.168.1.30 is at 00:0c:29:34:e5:ad	Reply from the client machine
11	Who has 192.168.1.1? Tell 192.168.1.30	Request in the other direction
12	192.168.1.1 is at 00:50:56:c0:00:01	Reply from my computer

Frames 3 to 10 are missing because the display filter hides everything that is not ARP. They are the 4 ping requests and 4 replies (8 frames).

Error log

Mistake	Correct answer	Why
TTL 117: said 12 routers	11	128 - 117 = 11. I made an arithmetic slip.
Destination of the ARP	`ff:ff:ff:ff:ff:ff` (Broadcast)	`c0:00:01` is the end

Mistake	Correct answer	Why
request: answered with `c0:00:01`		of the sender's own MAC. The destination is broadcast because the sender does not know the owner of the IP.
Why numbering jumps from 2 to 11	The filter hides other frames	My guess was close: the display filter only shows ARP. The hidden frames are the ping packets.

Main lessons

- To read an ARP line: who asks (`Tell`), who is asked (`Who has`), and who answers (`is at`).

- Source and destination are different things. The sender's MAC is not the destination's MAC.
- `TTL=64` does not prove Linux. It only rules out Windows.
- A display filter hides frames but does not delete them. The numbering keeps counting everything captured.

Anki

Cards for: `ff:ff:ff:ff:ff:ff` = broadcast MAC, the TTL values (128 and 64), and `00:0c:29` = VMware.

Next session

Public vs private addresses (RFC1918), APIPA and loopback.